Machine Learning Project Synopsis

1. Project Title: Predictive Maintenance of Turbofan Engines using Machine Learning Classification

Predicting whether an aircraft engine will fail within the next 30 operating cycles from its sensor readings (NASA C-MAPSS FD001).

2. Team Members:

Name	Roll No.	Contribution
[Student Name 1]	[Roll No.]	Data understanding and Exploratory Data Analysis (Notebooks 01–02)
[Student Name 2]	[Roll No.]	Labelling, feature derivation (rolling features), engine-wise split (Notebook 03)
[Student Name 3]	[Roll No.]	Classification models, regularization and hyperparameter tuning (Notebook 04)
[Student Name 4]	[Roll No.]	Ensemble models, performance analysis and threshold tuning (Notebook 05)

3. Problem Statement:

Aircraft engines wear out gradually. An unexpected failure is dangerous and expensive, while servicing on a fixed schedule throws away parts that still have useful life; in the NASA data used here, engine lifetimes range from 128 to 362 cycles, so no single schedule fits every engine. Every flight cycle, an engine records dozens of sensor readings (temperatures, pressures and speeds), and these readings slowly drift as components degrade. The problem is to use this sensor history to decide, for each engine at each cycle, whether it is likely to **fail within the next 30 cycles**, which leaves enough lead time to plan maintenance. The main difficulties are noisy readings, strongly correlated sensors, an imbalanced label (only 15% of training rows are "fails soon") and the need for a model that works on engines it has never seen.

4. Objective:

- Build a leakage-free data pipeline: explore the data (EDA), create the 30-cycle failure label, and derive rolling-window features that capture each engine's recent degradation trend.
- Train and compare classifiers: Logistic Regression and Decision Tree, controlled with L1/L2 and depth/leaf-size **regularization** and tuned with engine-grouped **GridSearchCV**, plus the **ensembles** Random Forest (bagging) and Gradient Boosting (boosting).
- Evaluate all models with imbalance-aware metrics on 100 unseen test engines, tune the decision threshold to catch more failures, and select and justify a final model for maintenance decisions.

5. Dataset:

- **Dataset name/source:** NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) turbofan engine degradation data, subset **FD001** (one operating condition, one fault mode: high-pressure compressor degradation), published by the NASA Ames Prognostics Center of Excellence.
- **Link:** https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data
- **Approx. number of samples:** 33,727 rows (one row = one engine at one cycle): **20,631 training rows** from 100 engines run until failure, and **13,096 test rows** from 100 different engines stopped before failure (true remaining life given in RUL_FD001.txt).
- **Key features:** 21 sensor readings (temperatures, pressures, fan and core speeds, bleed flows) and 3 operating settings per cycle. After EDA, 10 constant columns are dropped and the 14 informative sensors are each extended with a 5-cycle **rolling mean** and **rolling standard deviation**, giving **42 model features**.

- **Target variable:** Binary label = 1 if Remaining Useful Life (RUL) ≤ 30 cycles ("fails within 30 cycles"), else 0, where RUL = last cycle of the engine − current cycle. Positive share: 15% of training rows, 2.5% of test rows.

6. ML Task & Proposed Approach:

- **ML task: Classification** (supervised, binary).
- **Proposed ML algorithms/models:** Logistic Regression (L1/L2 penalties), Decision Tree, Random Forest (bagging ensemble) and Gradient Boosting (HistGradientBoostingClassifier, boosting ensemble), all in scikit-learn.
- **How ML will be applied:** Each row is labelled from its remaining life. Constant columns are dropped and rolling features are computed per engine from the current and previous cycles only, so no future information leaks in. Models run inside scikit-learn Pipelines with standard scaling and are tuned by GridSearchCV with 5-fold **GroupKFold** grouped by engine, so validation engines are never seen in training. The tuned models are then compared on NASA's separate test engines; the best model's decision threshold is lowered using out-of-fold training predictions to raise recall, and a final "maintenance now?" decision is evaluated for each of the 100 test engines.

7. Evaluation Metrics:

Recall (primary: share of failing engines caught, since a missed failure costs far more than an extra inspection), **Precision**, **F1-score** (used for model selection and tuning), **ROC-AUC** and **PR-AUC** (threshold-independent; PR-AUC focuses on the rare positive class) and the **confusion matrix**. Overfitting is checked through the train vs. validation gap, and recall/precision are also reported per engine (one decision per test engine). Accuracy is reported but not used to compare models: always predicting "healthy" already scores 97.5% on the test set.

8. Expected Outcome:

A tuned classifier that flags engines within 30 cycles of failure, showing that derived rolling features, regularization and ensembling produce a reliable and explainable early-warning model. In our implementation the **Random Forest** gave the best cross-validated F1 (**0.871**) and test ROC-AUC / PR-AUC of **0.991 / 0.807**. With its threshold lowered to 0.33, it flagged 18 of the 25 test engines that needed maintenance, with only one false alarm in 19 flags, and caught every engine with fewer than 18 cycles left.

9. References / Dataset Link:

[1] A. Saxena, K. Goebel, D. Simon and N. Eklund, "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation," Proc. International Conference on Prognostics and Health Management (PHM), 2008.

[2] NASA Prognostics Center of Excellence, C-MAPSS Jet Engine Simulated Data: https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data

[3] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.